

CMAN-3

TASKS:

1. Given, $N \times N$ matrix
where $N = 32$ FP32 matrixes

$$C = A \times B$$

$$C[i][j] = \sum_k A[i][k] \times B[k][j]$$

$$A = M \times K \times N = 32 \times 32 \times 32 = 32,768 \times 4 = 131,072$$

$$B = M \times K \times N = 32 \times 32 \times 32 = 32,768 \times 4 = 131,072$$

$$\text{Here, } A = N^3 \times 4,$$

$$B = N^3 \times 4$$

$$\Rightarrow 2 N^3 \times 4$$

$$= 2 \times (32)^3 \times 4$$

$$= 262,144$$

$$= 256 \text{ KB}$$

=

$$\text{Writes} = N^2 \times 4$$

$$\text{Reads} = 2 N^3 \times 4$$

2. Given tile size $T = 8$

~~The total dimensions per~~

$$\text{No. of tiles per dimension} = \frac{N}{T} = \frac{32}{8} = 4$$

A tile uses across 4-j tiles

B tile uses across 4-i tiles.

Total 64 element each for A tile and B tile

No. of tile loads will be $\left(\frac{N}{T}\right)^3 = \frac{32^3}{8^3} = \frac{32768}{512}$

$$\Rightarrow \left(\frac{32}{8}\right)^3 = (4)^3 = 64$$

For A tile = $64 \times 64 = 4096$ elements $\left(\frac{N}{T}\right)^3 \times T^2$

B tile = $64 \times 64 = 4096$ elements $\left(\frac{N}{T}\right)^3 \times T^2$

Both A & B together 8192 elements.

$$2 \times \left(\frac{N}{T}\right)^3 \times T^2$$

$$\Rightarrow 8192 \times 4 \quad (\text{Total DRAM traffic})$$

$$\Rightarrow 32,768$$

$$= 32 \text{ KB}$$

3. The ratio of Naive DRAM traffic to tiled DRAM traffic is $\Rightarrow \frac{\text{Naive DRAM traffic}}{\text{tiled DRAM traffic}}$

$$\Rightarrow \frac{262,144}{32,768} \Rightarrow 8T$$

Tiling reduces DRAM traffic because each 8×8 block of A & B is reused across multiple

output computations instead of being repeated repeatedly.

As a result, total memory access drop from N^3 behaviour in naïve case to much fewer tile based loads, giving an 8x reduction in traffic.

4. Given Bandwidth = 320 GB/s
compute is 10 TFLOPS

For Naïve case

DRAM traffic = 256 KB

$$\begin{aligned} \text{Time} &= \frac{262,144}{320 \times 10^9} \\ &\Rightarrow 8.192 \times 10^{-7} \\ &\Rightarrow 0.819 \mu\text{s} \end{aligned}$$

compute time 0.007 μs .

Bottle neck memory Bound.

So, execution time is 0.819 μs .

For tiled case,

$$\begin{aligned}\text{DRAM traffic } 32 \text{ KB} \\ = 32,768 \text{ Bytes}\end{aligned}$$

$$\begin{aligned}\text{Bandwidth} &= 320 \text{ GB/s} \\ &= 320 \times 10^9 \text{ Bytes/s.}\end{aligned}$$

$$\text{Time} = \text{traffic} / \text{Bandwidth}$$

$$\begin{aligned}t &= \frac{32,768}{320 \times 10^9} \\ &\Rightarrow 1.024 \times 10^{-7} \text{ s} \\ &\Rightarrow 0.1024 \text{ }\mu\text{s}\end{aligned}$$

Compute time 0.007 μs

Bottle neck memory bound (but much improved ^(vs) than naive)

So, execution time is 0.1024 μs .

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